Lecture 7(d): Flexibility quantification for heat pumps

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Agenda

- Heat pump growth in distribution networks
- Flexibility quantification method for HP
- Mathematical model for FQ
- Numerical case studies
- Conclusions

Heat pump growth in distribution networks

Context

- According to the European Heat Pump Association, 25.5 million HPs were operating across 19 countries by the end of 2024, with 2.31 million sold in this year
- EU's REPowerEU plan targets 60 million installations by 2030

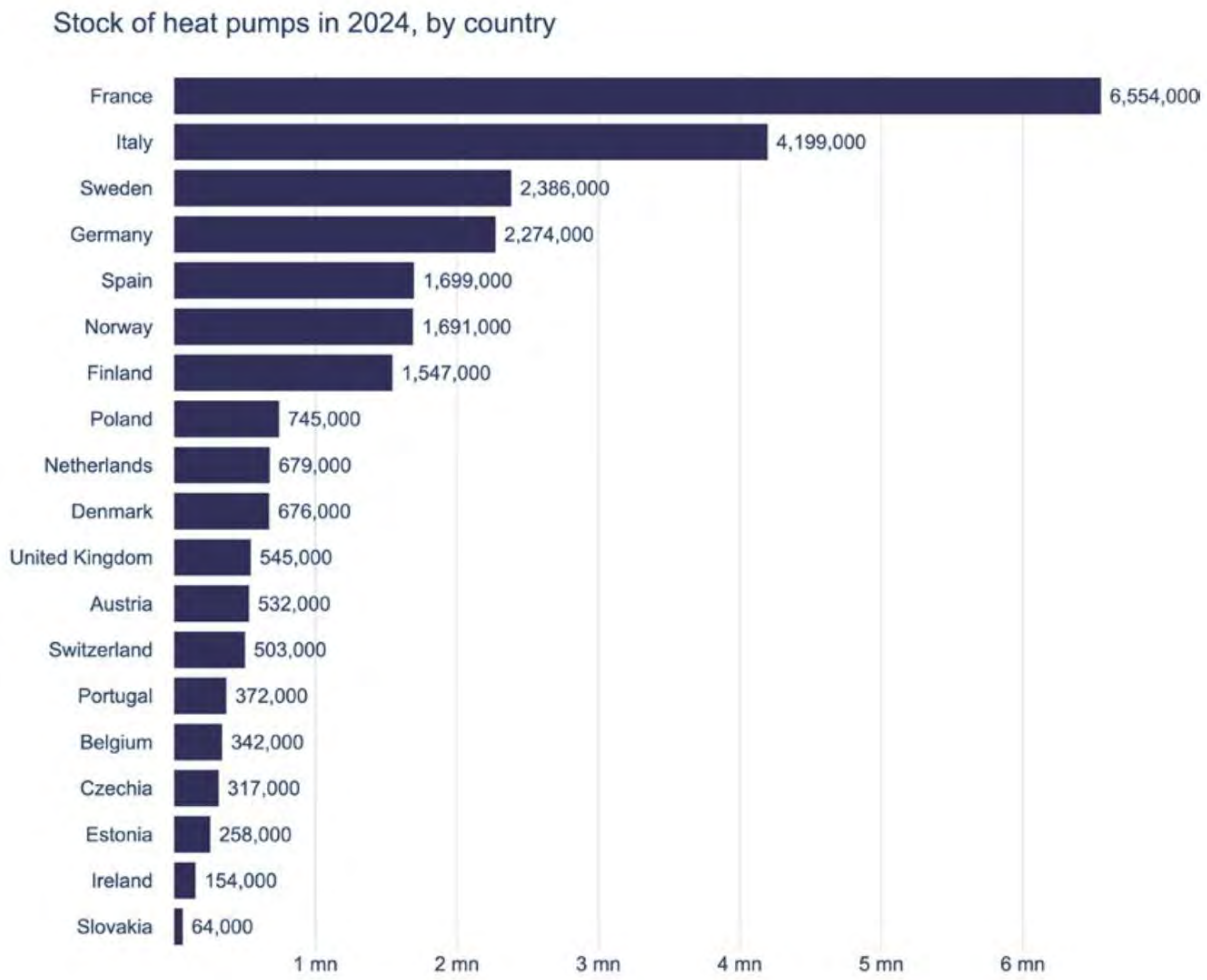
Shifting space heating to electricity moves the winter peak from gas to power

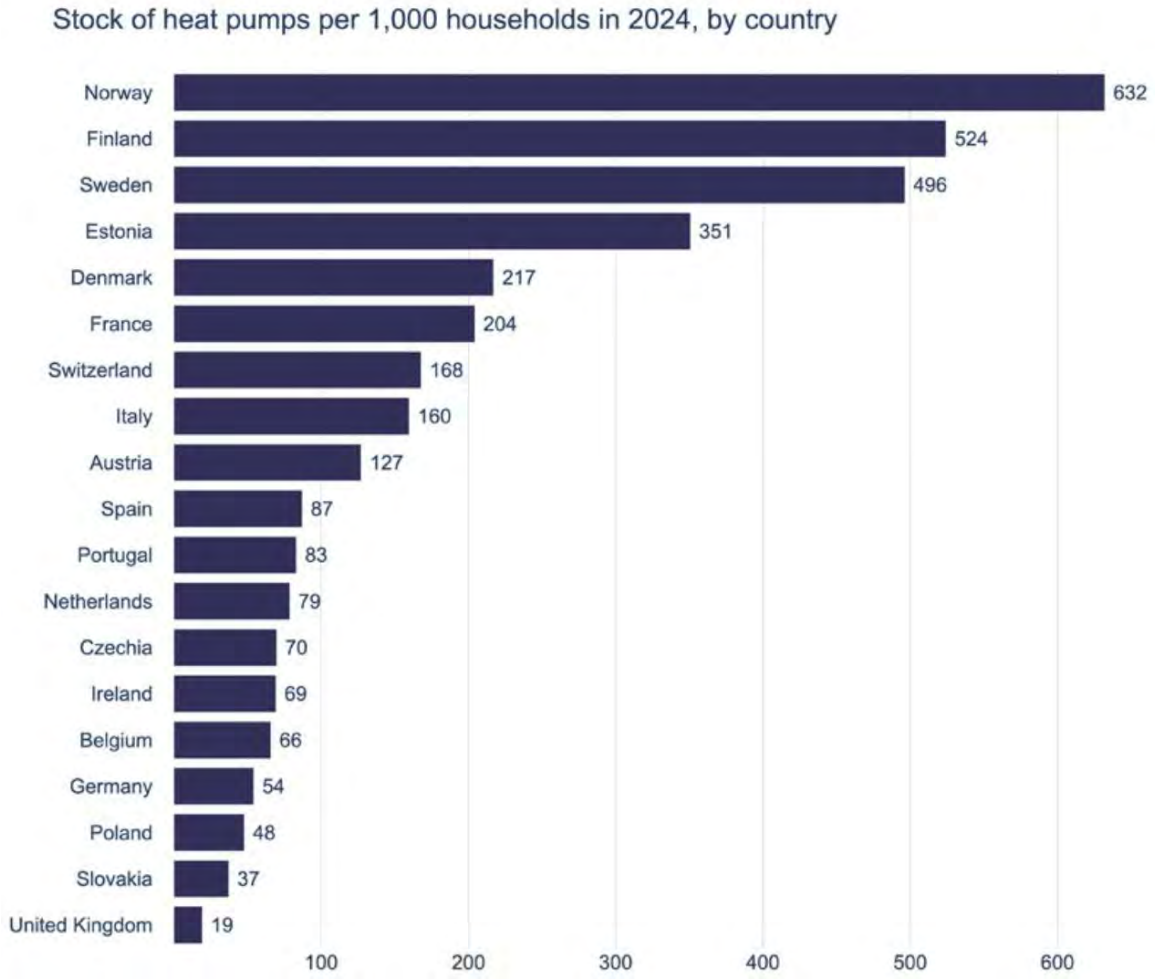
HP adoption could raise grid peak load by up to 70% without control

Sources:
[1] https://www.ehpa.org/wp-content/uploads/2025/07/EHPA-Market-Report-2025-executive-summary.pdf
[2] https://ehpa.org/news-and-resources/news/repowereu-heat-pump-strategy-required-to-help-sector-deliver-3/
[3] M. Waite and V. Modi, "Electricity load implications of space heating decarbonization pathways," Joule, vol. 4, no. 2, pp.376–394, 2020.

Heat pump landscape

Cumulative numbers by the year 2024



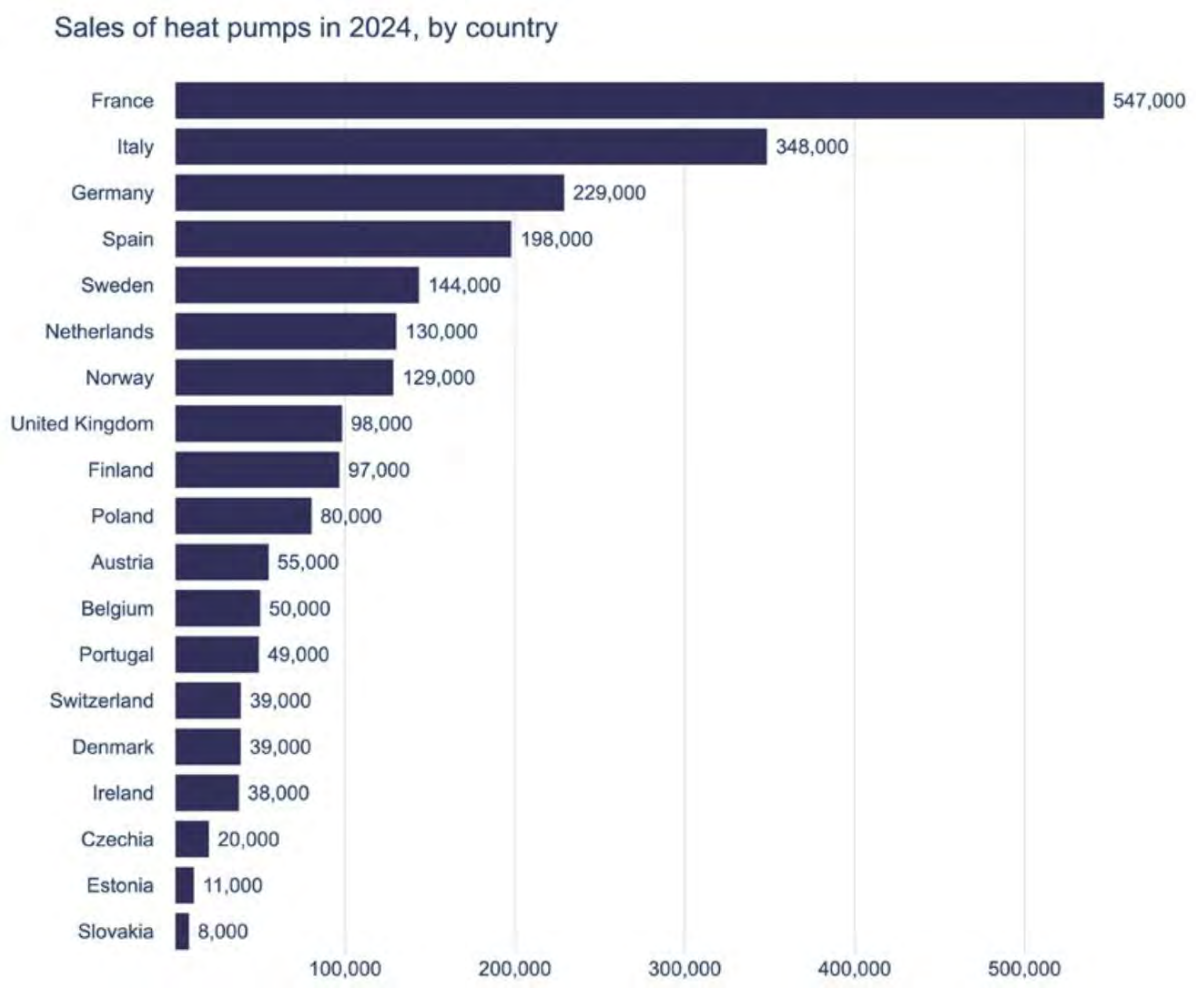


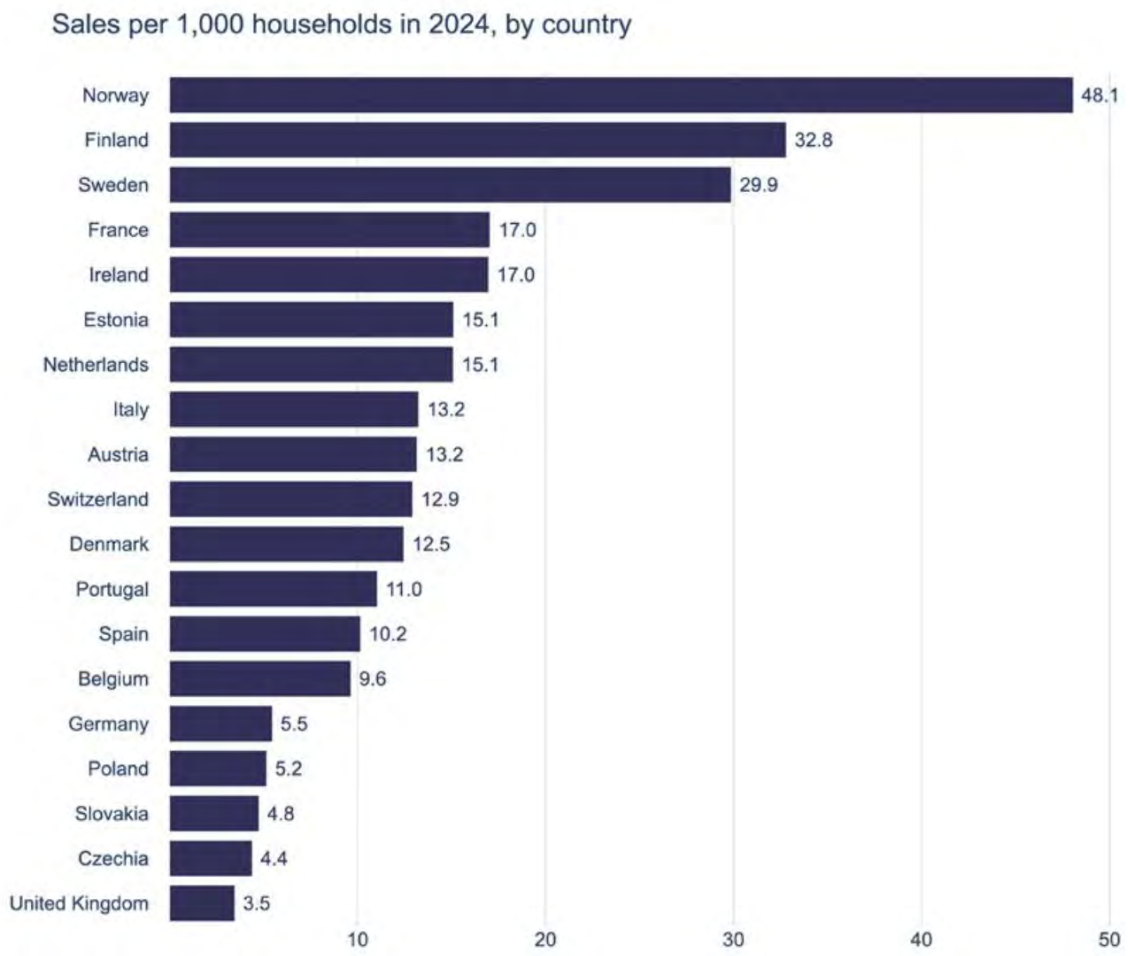
Sources:

[1] https://www.ehpa.org/wp-content/uploads/2025/07/EHPA-Market-Report-2025-executive-summary.pdf

Heat pump landscape

For the year 2024

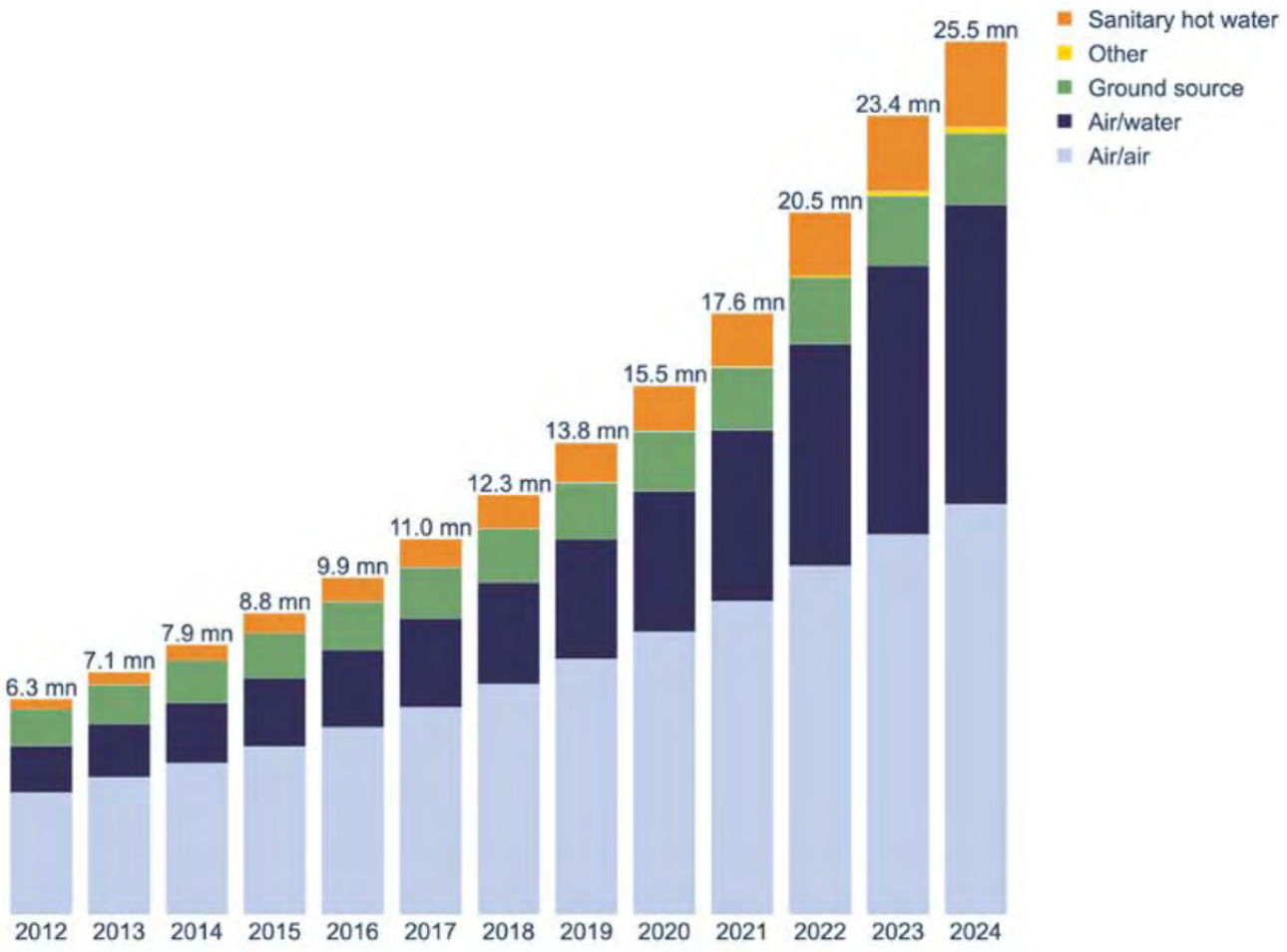




Sources:

[1] https://www.ehpa.org/wp-content/uploads/2025/07/EHPA-Market-Report-2025-executive-summary.pdf

Annual stock of heat pumps in 19 EU countries

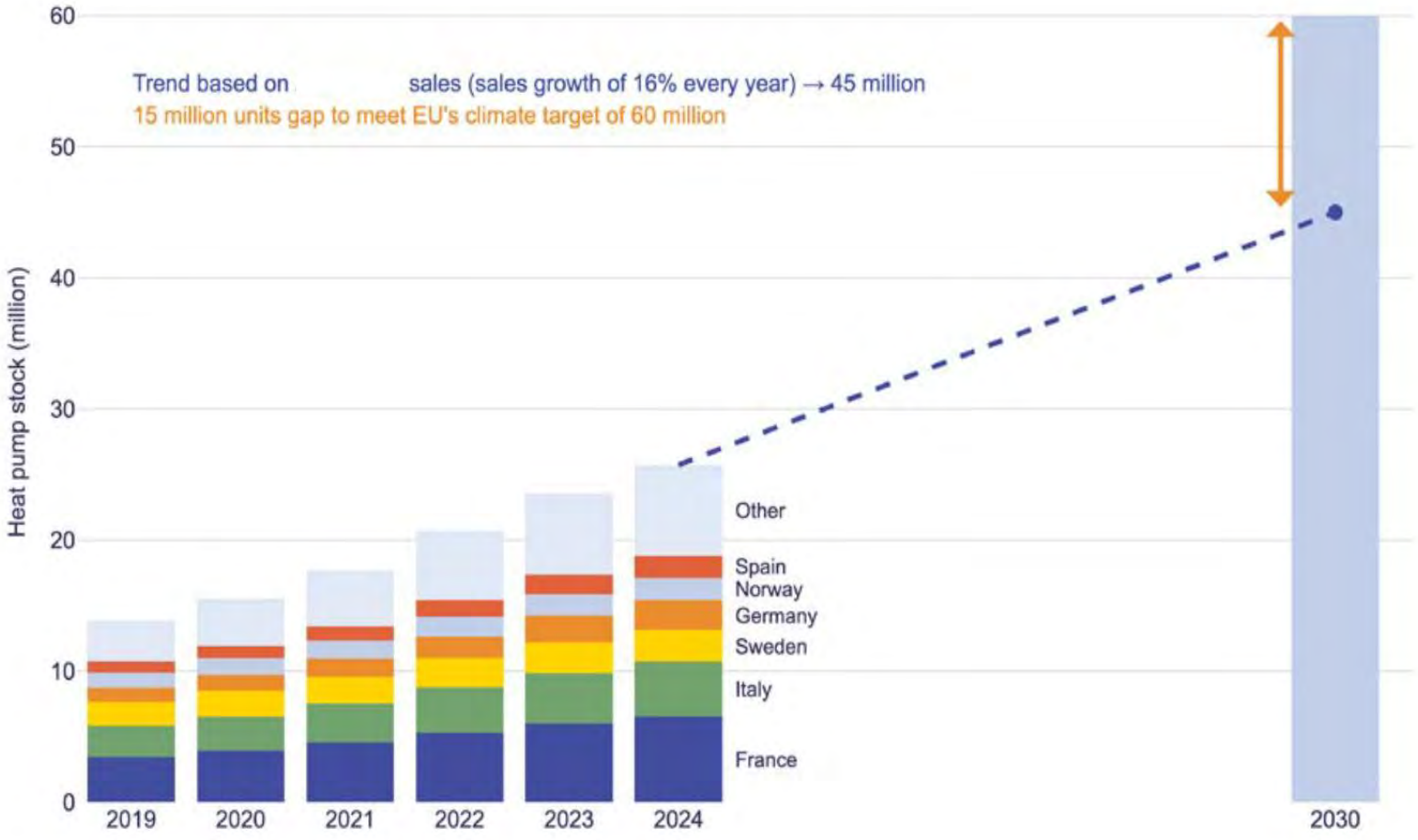


19 countries include: Austria, Belgium, Czechia, Denmark, Estonia, Finland, France, Germany, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Slovakia, Spain, Sweden, Switzerland and the United Kingdom.

What to expect by 2030?

In 2022, following the energy crisis and Russia's invasion of Ukraine, the EU set an objective that Europe should add 60 million more heat pumps by 2030 to rapidly reduce gas consumption in buildings.

REPowerEU-driven target to deploy 60 million additional heat pumps by 2030, beyond the ~20 million units already installed in 2022.



How does comfort levels affect power consumption?

- In Belgium, simulations for 2040 show that switching off units at an 18°C comfort threshold could cut 2.5 GW instantly and sustain 1.7 GW of relief for four hours, showing HP control can deliver gigawatt-scale peak shaving

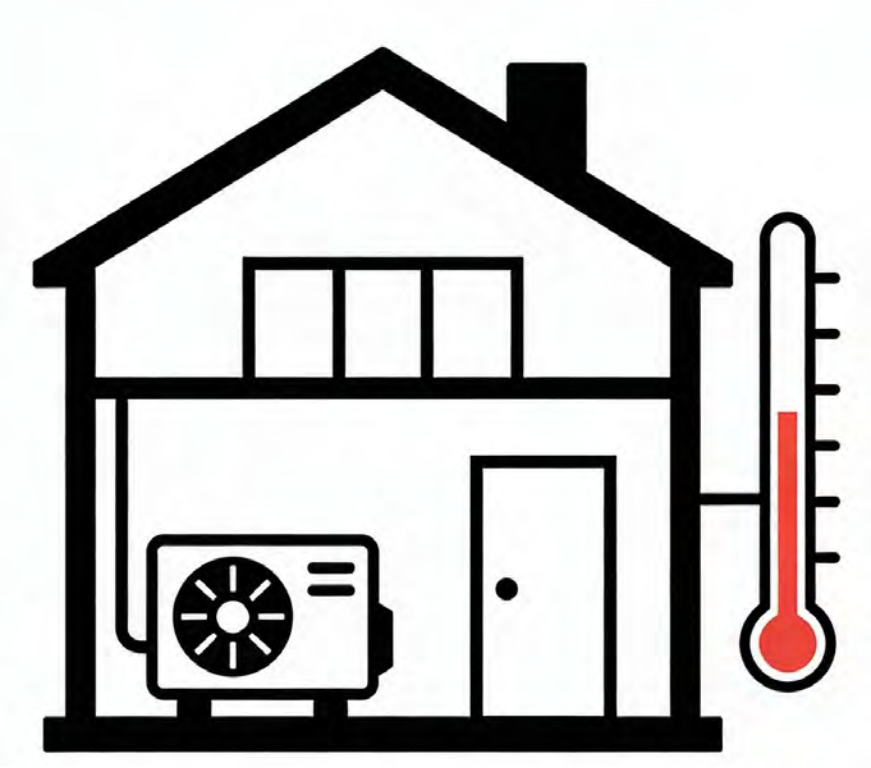
W. Bracke, “Flexibility potential of residential heat pumps in the power grid of the future,” ASHRAE Transactions, vol. 130,pp. 227–235, 2024.

Flexibility quantification method for HP

Notion of flexibility for HPs

Notion of flexibility for HPs: The notion of flexibility for heat pumps (or any other thermostatically controlled devices) relies on the flexibility in the comfort band of permissible temperatures. Greater flexibility potential is present in a larger thermal comfort band compared to a smaller band. There is no potential for flexibility if the comfort temperature range is a set point. HP flexibility relies on building characteristics, operating limits of the HP and the comfort band.

Flexibility potential of heat pump



The flexibility potential of heat pump depends on

- Outdoor temperature
- Indoor temperature
- **Comfort band** of temperature
  - The comfort band decides the permissible indoor temperature
  - Preheat / precooling of the house to provide max flexibility
- Heat pump parameters
  - Power consumption: deciding the rate of change of indoor temperature
  - Coefficient of performance of the heat pump

What is coefficient of performance of heat pump?

- The Coefficient of Performance (COP) of a heat pump is a measure of how efficiently it moves heat relative to the amount of electrical energy it consumes.

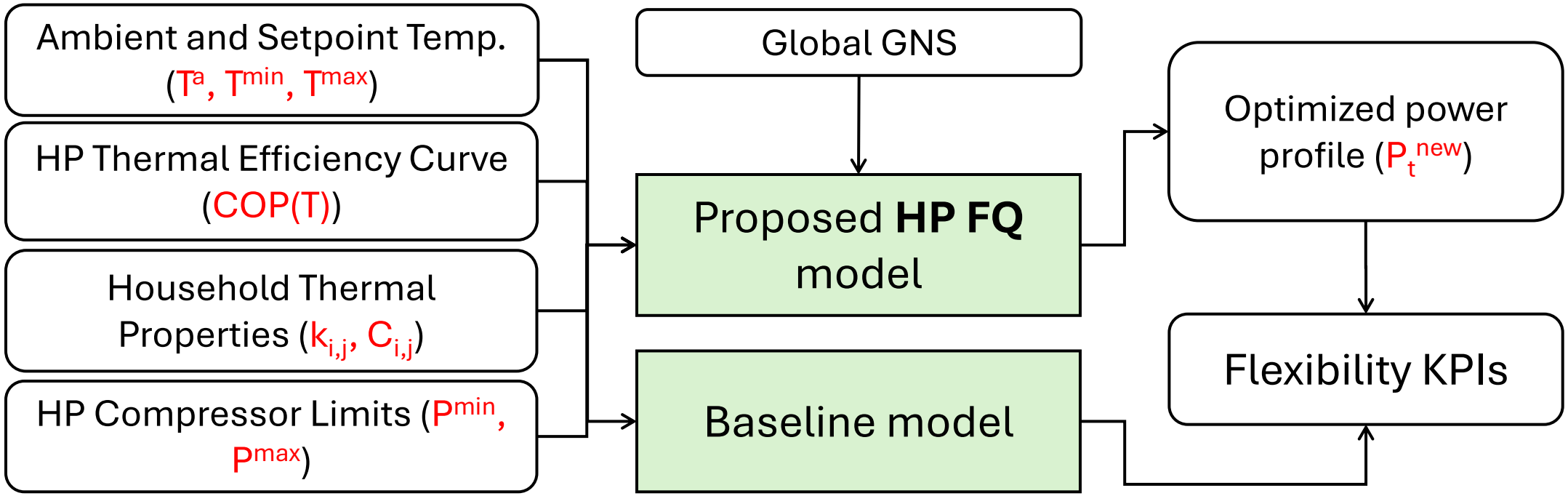
$$\text{COP} = \frac{\text{Heat output (useful heating delivered)}}{\text{Electrical energy input}}$$

- COP of 3 implies that for every 1 kWh of electricity the HP delivers 3 kWh of heat
- COP is typically greater than 1, because a heat pump does not generate heat—it moves ambient heat from outside (air, ground, or water) into the building

- Typical COP values
  - Air-source heat pumps: COP ≈ 2.5–4.0 (depending on outdoor temperature)
  - Ground-source heat pumps: COP ≈ 3.5–5.0

Mathematical model for FQ

FQ framework



Objective function for HP FQ

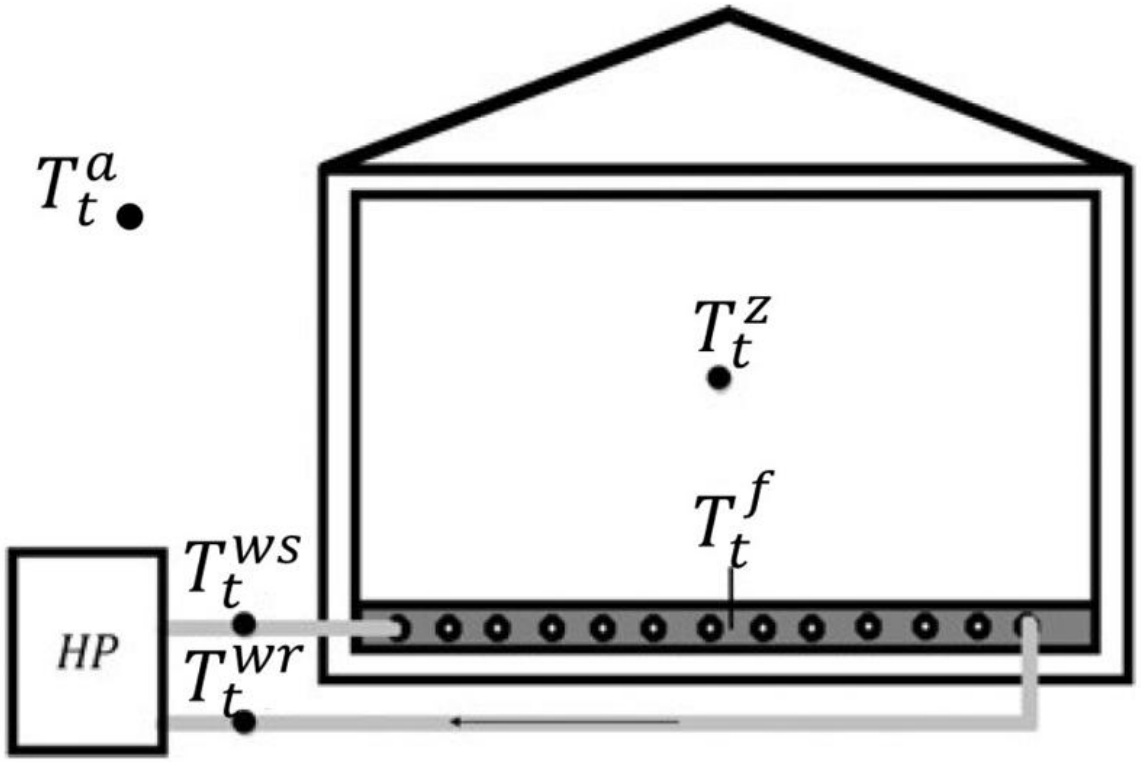
- Maximize revenue from providing flexibility

- Minimize penalty of indoor temperature deviation from the set value
  - Based on lower and upper bounds of the comfort level, the set value is selected as the mid point of these two levels
  - By ensuring target value of indoor temperature, we ensure that HP can provide ramp up or ramp down flexibility efficiently.

$$\max_{P_t^{\text{new}}} \Delta t \sum_{t \in T} \left(\lambda_t^{\text{up}} \delta_t^{+} + \lambda_t^{\text{down}} \delta_t^{-} - \lambda^{\text{penalty}} s_t \right)$$

HP modelling choices

- 1R-1C model (First-Order Model)
  - Represents the building as one thermal mass (e.g., indoor air + envelope lumped together).
- 2R-2C Model (Second-Order Model)
  - Captures the air and envelope
- 4R-4C Model (Higher-Order Model)
  - Captures Multi-layer envelope, internal mass, delays
- …



C. Verhelst, F. Logist, J. V. Impe, and L. Helsen, “Study of the optimal control problem formulation for modulating air-to-water heat pumps connected to a residential floor heating system,” Energy and Buildings, vol. 45, pp. 43–53, 2012

Numerical case studies

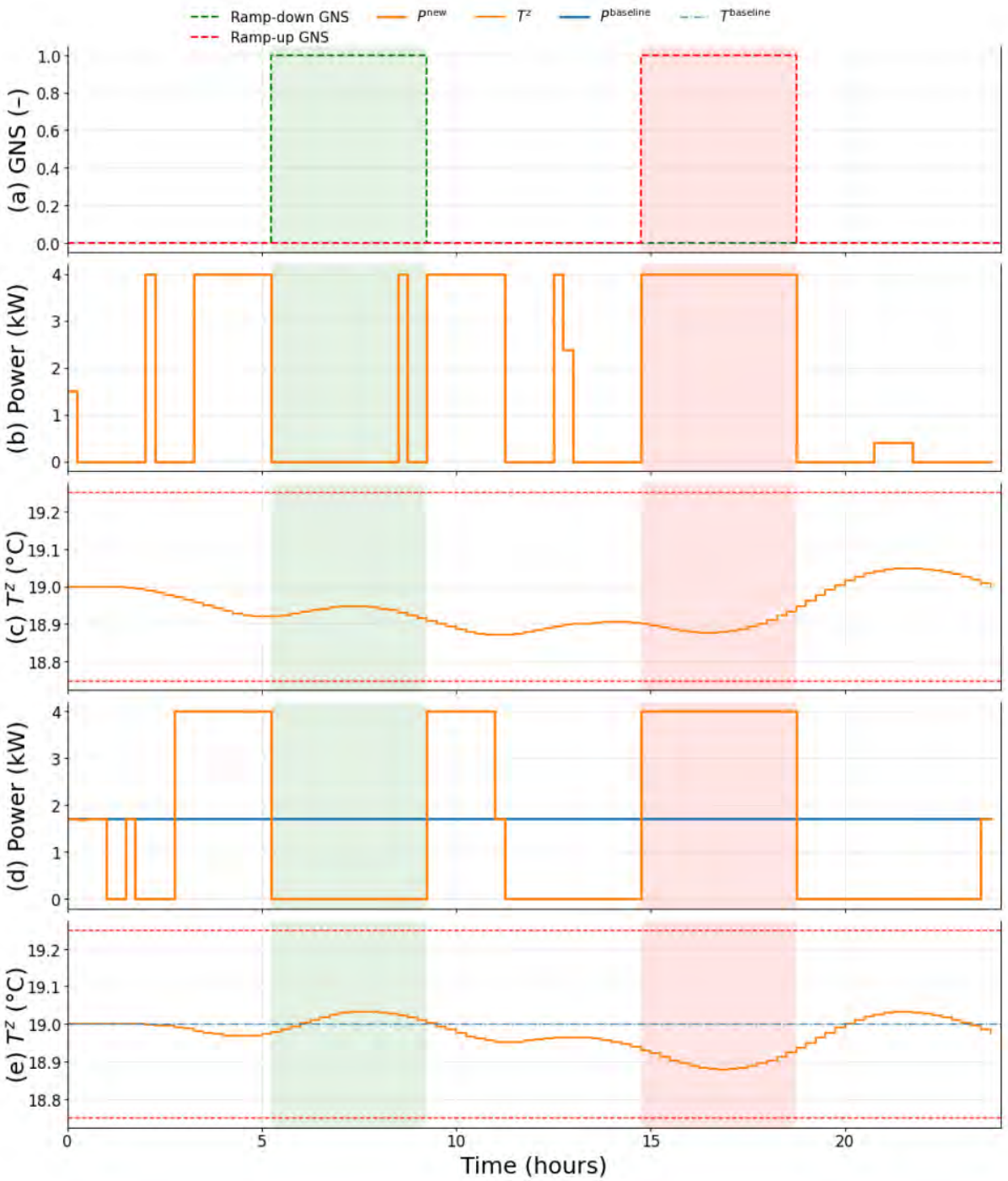
Case study: Exemplary case

- P_{min} = 0.4 kW, P_{max} = 4.0 kW
- Desired indoor temperature range: $T_z \in [18.75, 19.25]^{\circ}C$
- Outdoor temperature: $T_a = 0^{\circ}C$

Exemplary case

Exemplary HP FQ case with five aligned panels:

- (a) Global GNS
- (b) power profile, no baseline
  - (c) corresponding zone temperature, no baseline
- (d) power profile, with baseline
  - (e) corresponding zone temperature, with baseline



Results

Table VIII: KPI overview per scenario for the HP exemplary case

Base-line	E^{flex} [kWh]	E^{flex}_{up} [kWh]	E^{flex}_{down} [kWh]	ϵ^{abs}_{needed}	ϵ^{flex}_{needed}	β^{flex}	$\epsilon^{flex}_{\lambda,max}$	n_{cycle}
No	16.0	16.0	0.0	0.89	0.39	0.17	1.0	/
Yes	16.0	9.18	6.82	0.89	0.39	0.17	1.0	/

Does baseline have impact on harvested flexibility?

Conclusions

Conclusions

- The flexibility provided by heat pump depends on
  - Comfort band
  - Outdoor temperature
  - Heat pump parameters
- Thermal inertia and consumer flexibility reflected in terms of comfort band has a large potential for grid services

Thank you for your attention!

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